



Intelligent Academic Path Optimizer

- *Anndeen Daley – Data Engineer/Database*
- *Bella Gerken – AI/NLP Integration*
- *Charles Varga – DevOpS /Tech Architect*
- *James Simpson – PO/SCRUM Master*
- *Tara Edney – Frontend*
- *Younes Roumila - Backend*

Agenda



Product
Description



Technologies



Minimum
Viable
Product



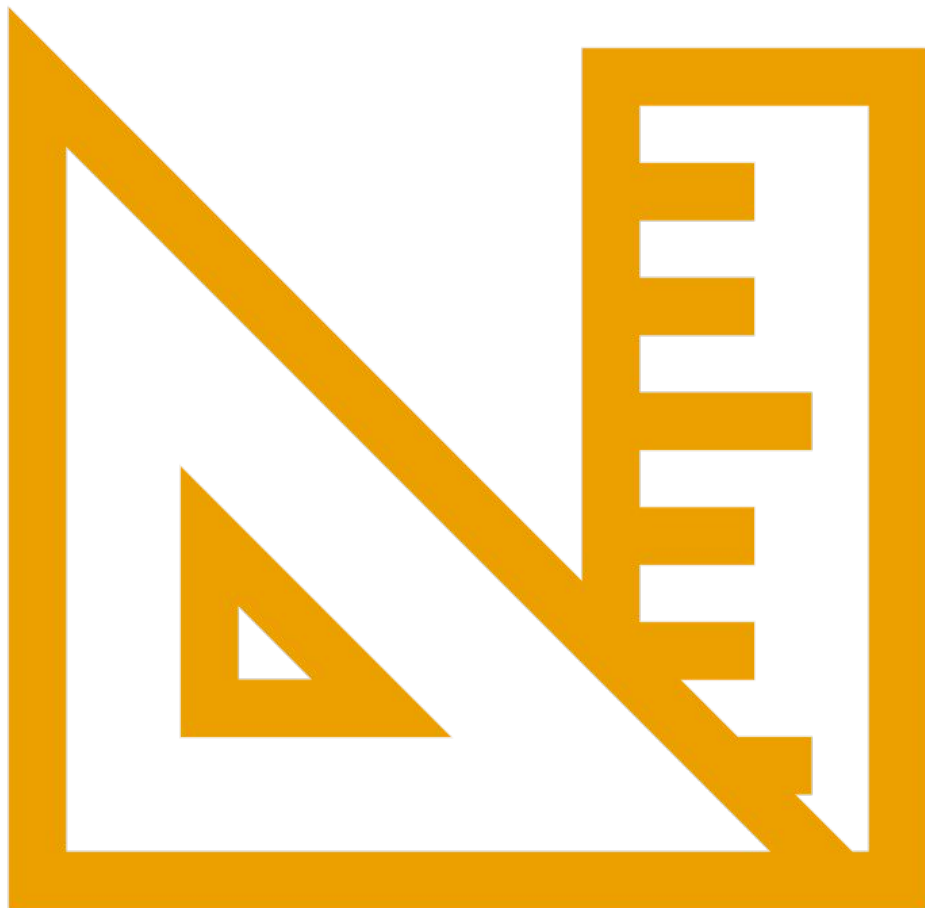
Personas



Sprint
Schedules



Retrospective



Project Description

The Intelligent Academic Path Optimizer is a proposed application which will streamline the academic advising workflow. The app is designed to use AI to facilitate decision making when generating a conflict-free academic schedule for students. By utilizing the Constraint Satisfaction Problem algorithms and predictive modeling, the application will dynamically output the most efficient path to graduation based on specific student parameters.

Tech Stack

Frontend (React Website)

- Figma
- React JS
- Native CSS
- Email JS

Backend (Web Server)

- Python 3.12 w.
FastAPI
- Gemini API
(AI Chat)
- AWS
DynamoDB + S3
(Accounts)

Web Hosting

- Github Pages
(Frontend)
- PythonAnywhere
(Web Server)
- AWS
(User Storage)
- Gemini (AI Server)

Minimum Viable Product (core features)

- A user-friendly interface in which the user can interact with the AI tied to the app to receive feedback based on the input
- A text input in which students can specify their particular academic needs to the AI
- A clearly visible feedback mechanism which users can utilize to inform developers about the AI, particularly in cases when it generates incorrect information or hallucinates
- A means for users to select the major that they are pursuing and at which level, be it undergraduate or graduate
- Output from the AI which consists of the classes that the students should take based on specified major as well as any text input for specific academic needs in the prompt, and in which semesters these students should take specific classes

Personas

Academic Advisor: Isabelle Johnson, 42 · Pace University

Problem

- Manually verifies prereqs and builds schedules for 100+ students
- Changing requirements risk graduation delays

Goals

- Accurate prereq and degree checks
- Automated scheduling
- Earlier risk flags
- More time to help students

How IAPO Helps

- Generates valid pathways so she can focus on mentorship

Graduate Student: William Court, 24 · CS Major

Problem

- Confused about prereqs and availability each semester
- Unclear requirements delay graduation and raise costs

Goals

- Fastest path to graduate
- "What-if" planning before registering
- Balanced schedule
- Steady GPA

How IAPO Helps

- Builds an optimized schedule from his credits, goal date, and load

Dept. Administrator: Michael Rodriguez, 52 · CS Chair

Problem

- Little visibility into which prereqs cause delays
- Relies on manual progress reviews

Goals

- Spot bottlenecks
- Track student progression
- Support data-driven decisions
- Raise grad rates

How IAPO Helps

- Surfaces bottlenecks from degree plans to guide curriculum planning

Sprint Schedule

Sprint 0 May 21 - June 14					
Name	Assigned To	Status	Priority	Estimation (days)	Notes
Minimum Viable Product Defined	Charles/Tara	Complete	High		
Create User Personas	Bella/Andeen	In Progress	High		
Create Team Roles	Team	Scheduled – meeting			Scheduled for June 9 meeting
Establish Weekly Meetings	Team	Scheduled – meeting			Scheduled for June 9 meeting
Set up Github and Wikipage	Charles	Complete	High		
Deliverable 1 - Put Together in PPT & video	James	Scheduled			
Create Schedule	Younes/James	Complete	High		
Create User Stories	Team				Use extra time to work ahead before end of sprint 0
Sprint 1 June 14 - July 13					
Name	Assigned To	Status	Priority	Estimation (days)	Notes
Design App	Frontend				
Create Homepage	Frontend				
Setup Email JS	Frontend				Need backend coordination
Stylize Website	Frontend				
Deploy to Webserver	Frontend				
Create python webserver	Backend				
Connect to AWS S3	Backend				
Implement User Account Creation	Backend				
Set up AI integration	Backend				
Create Agile Board	SCRUM Master				
Sprint 1 Deliverable	Team				
Sprint 2 July 14 - Aug 12					
Name	Assigned To	Status	Priority	Estimation (days)	Notes
Connect Frontend to Backend					
Anything left over					General Placeholder

Retrospective

- Tasks/Stories complete:
 - MVP defined
 - Create User Personas
 - Create Team Roles
 - Establish Weekly Meetings
 - Set Up Github and Wikipage
 - Create Schedule
- Tasks/Stories incomplete:
 - Final video edited/uploaded – James (can't be done until this meeting concludes)
 - Create User stories – Using extra time from June 9 – 14 to generate these before Sprint 1
- Metrics
 - 6/8 tasks complete
 - Final video can't be complete until after meeting.
 - Extra time in Sprint 0 to get a head start on Sprint 1 User Stories

What went well?

What needs improvement?